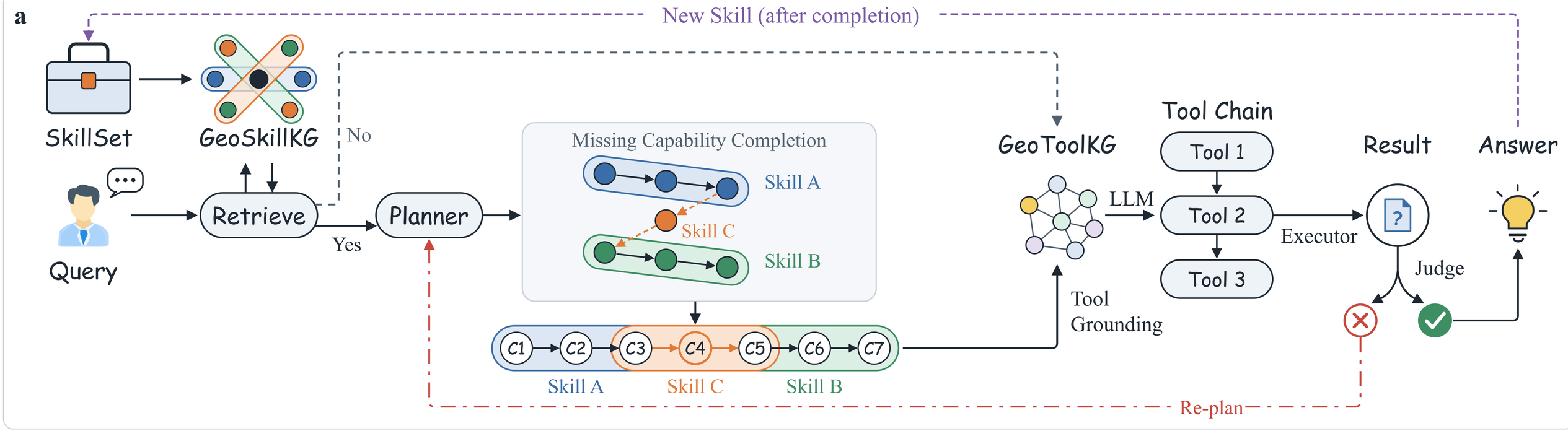
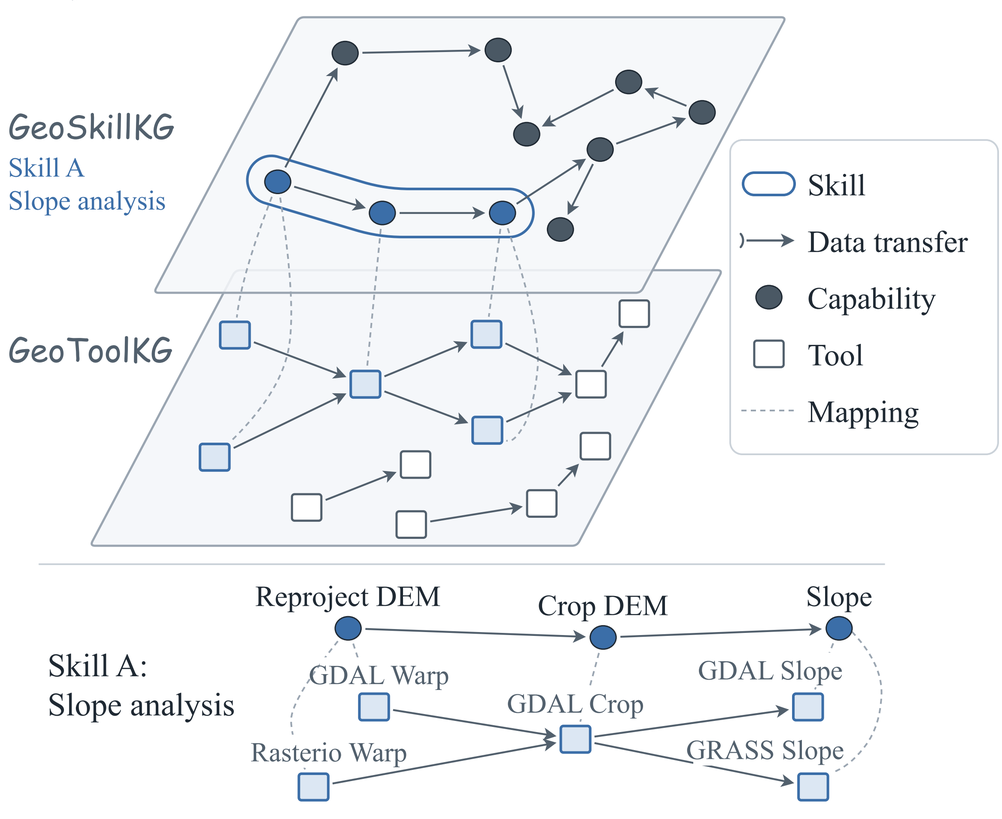
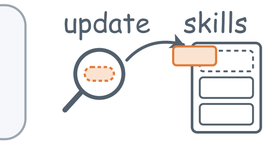
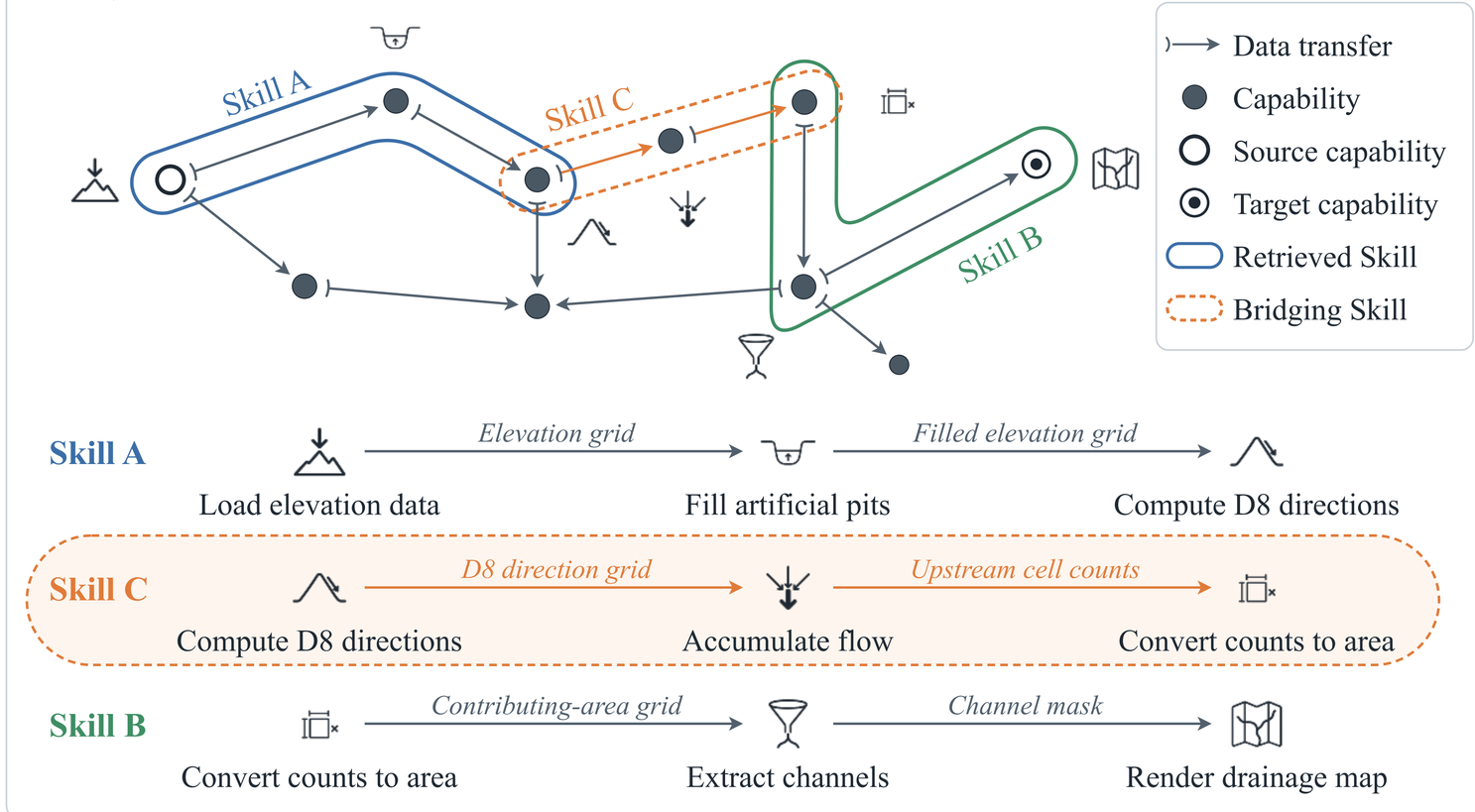

a
New Skill (after completion)
SkillSet
GeoSkillKG
No
Query
Retrieve
Yes
Planner
Missing Capability Completion
Skill A
Skill C
Skill B
C1
C2
C3
C4
C5
C6
C7
Skill A
Skill C
Skill B
GeoToolKG
Tool Grounding
LLM
Tool Chain
Tool 1
Tool 2
Tool 3
Executor
Result
Judge
Answer
Re-plan


b
Reproject this elevation map to a local metric CRS, crop it to the study-area extent, and calculate slope in degrees.
GeoSkillKG
Skill A
Slope analysis
GeoToolKG
Skill
Data transfer
Capability
Tool
Mapping
Skill A:
Slope analysis
Reproject DEM
Crop DEM
Slope
GDAL Warp
GDAL Crop
GDAL Slope
Rasterio Warp
GRASS Slope


update
skills


c
Fill artificial pits in this elevation map. Map channels with ≥1 km² upstream area; report slope, flow-direction and coverage distributions, plus the channel-cell count.
Skill A
Skill C
Skill B
Data transfer
Capability
Source capability
Target capability
Retrieved Skill
Bridging Skill
Skill A
Elevation grid
Filled elevation grid
Load elevation data
Fill artificial pits
Compute D8 directions
Skill C
D8 direction grid
Upstream cell counts
Compute D8 directions
Accumulate flow
Convert counts to area
Skill B
Contributing-area grid
Channel mask
Convert counts to area
Extract channels
Render drainage map